

Computer Science Fundamentals

Channel Coding – CRC Codes

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- Linear, cyclic block codes
 - **cyclic**: circular shifts of a code word result in a valid code word
- Design goals:
 - **Detection** of
 - Single- and double-bit errors
 - Burst errors (several erroneous bits in a row)
 - easy implementation (especially in hardware)
- Used, e.g.
 - Ethernet, USB, Bluetooth, SCSI, Serial ATA, ISDN, DECT (cordless phones), CAN, FlexRay (Automotive)
 - ...

- Attach a k bit CRC code to an n bit long message
- Interpret message as **coefficients of a dyadic polynomial**
 - **Dyadic** = calculate modulo 2 (thus, coefficients can only assume values 0 and 1)
 - CRC is based on polynomial division
- Example
 - Message: 10011010
 - Polynomial $N(x) =$

$$1 \cdot x^7 + 0 \cdot x^6 + 0 \cdot x^5 + 1 \cdot x^4 + 1 \cdot x^3 + 0 \cdot x^2 + 1 \cdot x + 0 = x^7 + x^4 + x^3 + x$$

- Choose a **generator** polynomial $C(x)$ of degree k (k = length of attached CRC code)
- Transmit a polynomial $S(x)$
 - derived from $N(x)$
 - such that $S(x)$ is **divisible** by $C(x)$ **without remainder**
- Example $k = 3$
 - $C(x) = x^3 + x^2 + 1$
 - transmit: $S(x) = N(x) + k \text{ Bit}$

Steps

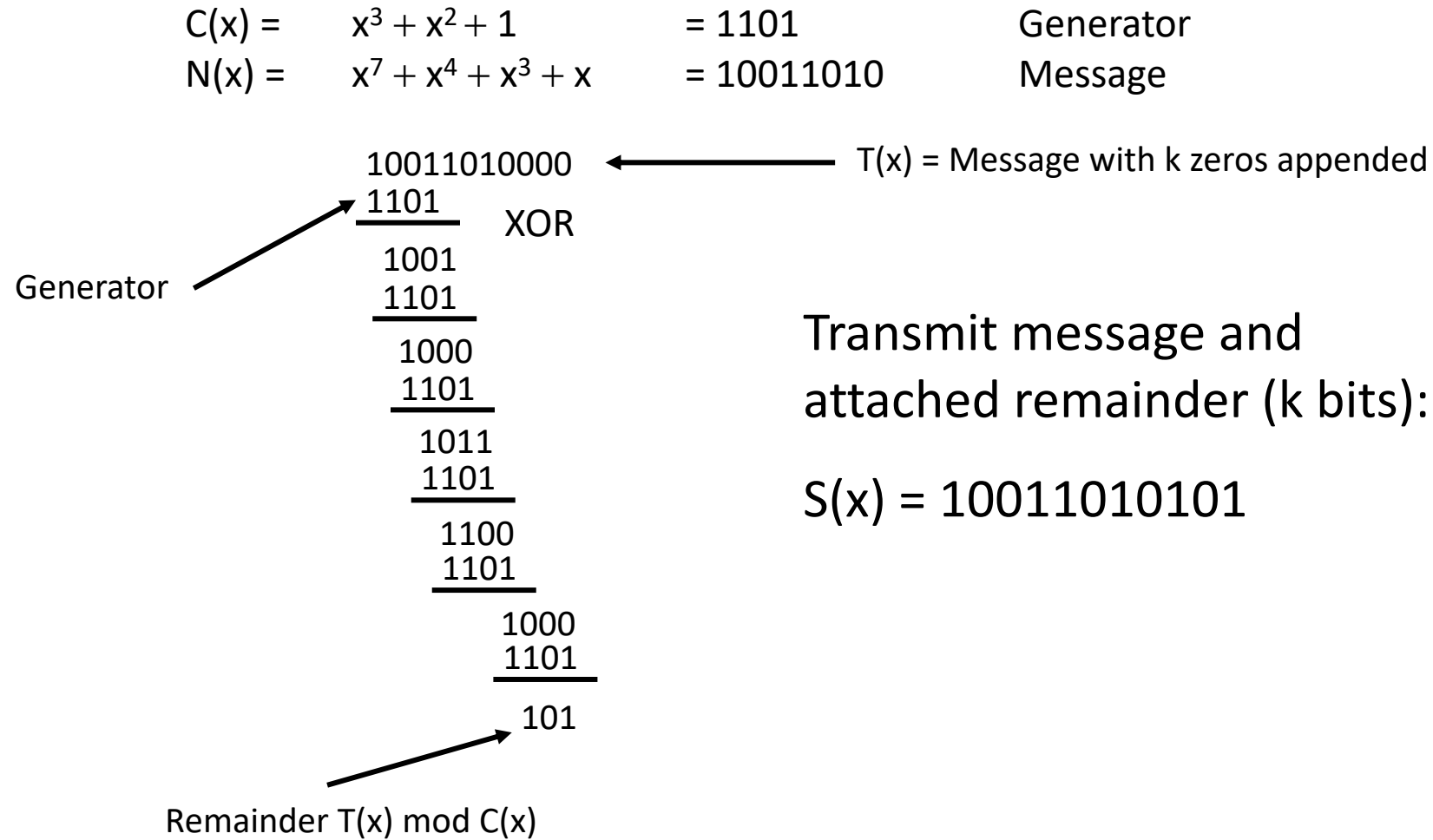
- $T(x) = N(x) \cdot x^k$ $\rightarrow$ append k zeros to message
- Calculate remainder $R(x)$ from division $T(x) / C(x) \rightarrow T(x) \bmod C(x)$
- Send $S(x) = T(x) - R(x)$
 - as we have mod 2 $\rightarrow T(x) - R(x) = T(x) + R(x)$
 - i.e.: append $R(x)$ to $N(x)$

Example

- | | | | | |
|------------|-------------|---|--------------------------------------|---------------------|
| • $N(x) =$ | 10011010 | = | $x^7 + x^4 + x^3 + x$ | |
| • $C(x) =$ | 1101 | = | $x^3 + x^2 + 1$ | $\rightarrow k = 3$ |
| • $T(x) =$ | 10011010000 | = | $x^{10} + x^7 + x^6 + x^4$ | |
| • $R(x) =$ | 101 | = | $x^2 + 1$ | |
| • $S(x) =$ | 10011010101 | = | $x^{10} + x^7 + x^6 + x^4 + x^2 + 1$ | |

- All calculations are mod 2
- Therefore, we have $1 + 1 = 1 - 1 = 0$
- Subtraction can be done by bitwise XOR of coefficients
- Always start with the leftmost coefficient of the message $N(x)$ (or rather its extension $T(x)$)

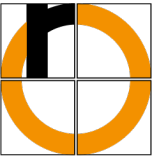
CRC – Polynomial Division – Example (Sender)



Steps

- Received polynomial $S'(x)$
 - Calculate remainder $R'(x)$ of division $S'(x) / C(x) \rightarrow S'(x) \bmod C(x)$
 - remainder = 0
 - error-free transmission
 - or undetectable error
 - remainder $\neq 0$
 - at least 1 bit in message is incorrect
 - message must be re-sent
- Note: error-correction is in principle possible (depending on generator), but rarely used with CRC

CRC – Example (Receiver, error-free)



$$\begin{array}{llll} C(x) = & x^3 + x^2 + 1 & = 1101 & \text{Generator} \\ S'(x) = & x^{10} + x^7 + x^6 + x^4 + x^2 + 1 & = 10011010101 & \text{Received message} \end{array}$$

Generator $\swarrow$

$$\begin{array}{r} 10011010101 \\ \underline{1101} \quad \text{XOR} \\ 1001 \\ \underline{1101} \\ 1000 \\ \underline{1101} \\ 1011 \\ \underline{1101} \\ 1100 \\ \underline{1101} \\ 1101 \\ \underline{1101} \\ 0 \end{array}$$

$\swarrow$ Remainder $S'(x) \bmod C(x)$

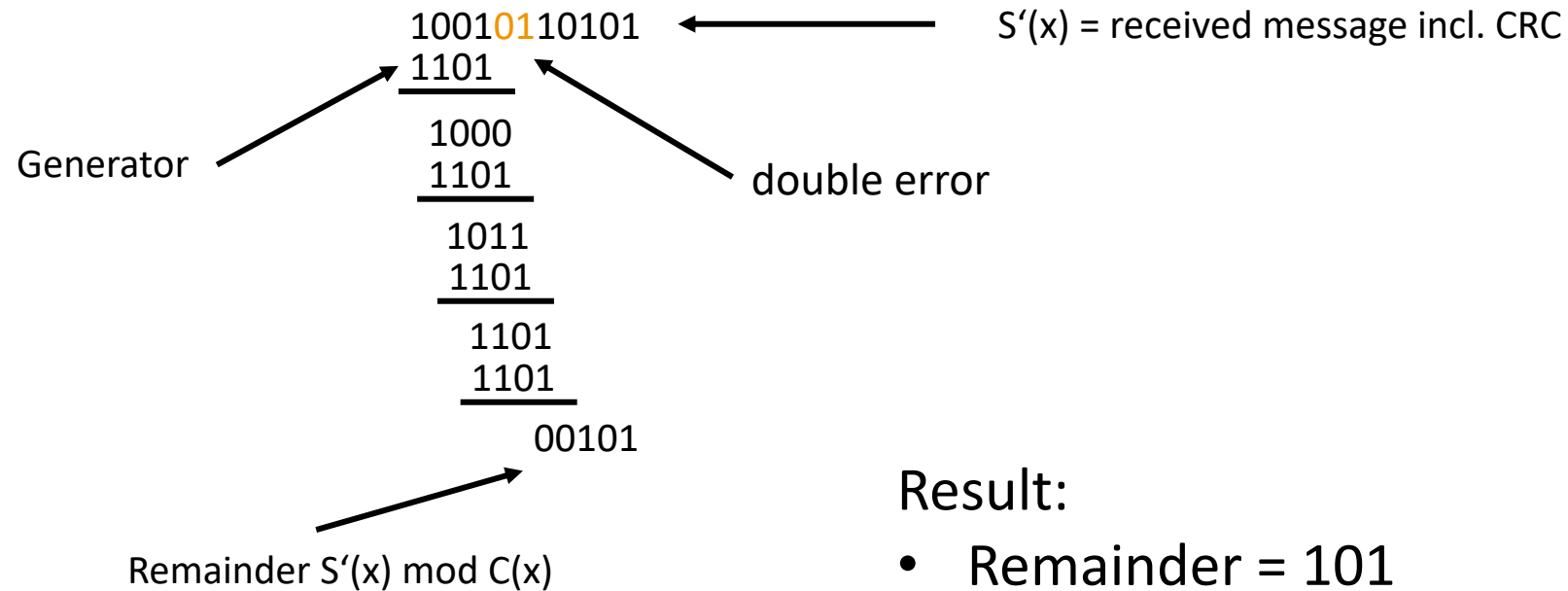
$\longleftarrow S'(x) = \text{received message incl. CRC}$

Result:

- Remainder = 0
- CRC check ok

CRC – Example (Receiver, with errors)

$$\begin{array}{llll} C(x) = & x^3 + x^2 + 1 & = 1101 & \text{Generator} \\ S'(x) = & x^{10} + x^7 + x^5 + x^4 + x^2 + 1 & = 1001\textcolor{brown}{0}110101 & \text{Received message} \end{array}$$

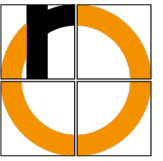


Result:

- Remainder = 101
- CRC check not ok

- Received polynomial $S'(x) = S(x) + F(x)$
 - $F(x)$ is a polynomial that represents the erroneous bits
 - $F(x) = 0 \rightarrow$ no errors
- All errors can be detected where $F(x)$ is not a multiple of $C(x)$
 $\rightarrow$ Requirements for generators $C(x)$
- Which errors can be detected?
 - **all single-bit errors**, if x^k and the constant term 1 exist
 - **all double errors**, if $C(x)$ has at least three terms, and the size of the data is smaller than the cycle length of $C(x)$
 - **all r -bit errors for odd r** , if $C(x)$ has an even number of terms; especially if it contains the factor $(x + 1)$
 - **all burst errors of length smaller k** , if $C(x)$ contains the constant term
 - **most** burst errors of length $\geq k$

CRC – Some Common Generator Polynomials



Name	Usage	Polynomial
CRC-1	Parity bit	$x + 1$
CRC-4-CCITT	Telecommunication = (15,11) Hamming	$x^4 + x + 1$
CRC-5-USB	USB	$x^5 + x^2 + 1$
CRC-5-Bluetooth	Bluetooth	$x^5 + x^4 + x^2 + 1 = (x^4 + x + 1)(x + 1)$
CRC-8-ITU-T	ISDN	$x^8 + x^2 + x + 1 = (x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + 1)(x + 1)$
CRC-15-CAN	CAN bus	$x^{15} + x^{14} + x^{10} + x^8 + x^7 + x^4 + x^3 + 1 = (x^7 + x^3 + x^2 + x + 1)(x^7 + x^3 + 1)(x + 1)$
CRC-32	Ethernet, Serial ATA, ...	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$

For protection during transmission, we want to use a CRC code.

The (binary) **message** to be sent is:

1100 0110

As **generator polynomial** we use:

$$x^6 + x + 1$$

What is the message to be sent, including the attached CRC code?